decode:VCE

STUDENT NUMBER | | | | | | | | | | Letter | |

CHEMISTRY

Written Examination

Trial Examination 1

Day Date

Reading time: *. to *.** (15 minutes)**
Writing time: *. to *.** (2 hours 30 minutes)**

QUESTION AND ANSWER BOOK

Structure of book

Section	*Number of questions*	*Number of questions to be answered*	*Number of marks*
A	30	30	30
B	10	10	90
			Total 120

- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners and rulers.
- Students are NOT permitted to bring into the examination room: blank sheets of paper and/or correction fluid/tape.
- No calculator is allowed in this examination.

Materials supplied

- Question and answer book of 28 pages.

Instructions

- Write your **student number** in the space provided on the front cover of the answer book.
- Answer all questions in the answer book.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.

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SECTION A - Multiple-choice questions

Instructions for Section A

Circle the response that is **correct** for the question.
A correct answer scores 1, an incorrect answer scores 0. Marks will **not** be deducted for incorrect answers.
No marks will be given if more than one answer is completed for any question.
Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

Question 1

Which of the following fuels can be considered carbon-neutral?

A. coal

B. biodiesel

C. coal seam gas

D. crude oil

Question 2

Which of the following species contains chromium in a different oxidation state to the other options?

A. CrO_4^{2-}

B. Cr_2O_3

C. $H_2Cr_2O_7$

D. CrO_3

Use the following information to answer Questions 3 and 4.

In 1999, a group of scientists reported the use of a solid oxide fuel cell that utilises methane as the fuel, which reacts directly at a ceria-based electrode. Oxygen reacts at the other electrode.

Question 3

Oxygen reacts at the

A. positive electrode, which is the anode.

B. positive electrode, which is the cathode.

C. negative electrode, which is the anode.

D. negative electrode, which is the cathode.

Question 4

The primary energy conversion in this fuel cell is

A. chemical to mechanical.

B. heat to chemical.

C. electrical to chemical.

D. chemical to electrical.

Question 5

Which of the following is the strongest **reductant**?

A. Fe^{2+} (aq)

B. H_2O (l)

C. H_2O (l)

D. Ni (s)

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Use the following information to answer Questions 6 and 7.

Hydrogen and iodine can be reacted reversibly at 400 °C as per the equation below:

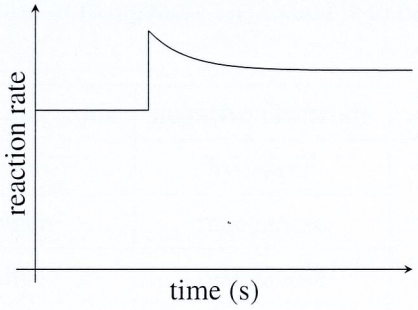
$$H_2\,(g) + I_2\,(g) \rightleftharpoons 2HI\,(g)$$

Given amounts of H_2 and I_2 were added to a 2.00 L reaction vessel, and equilibrium was established. The volume of the reaction vessel is then sharply decreased to 1.20 L at a particular time within the range of its reaction rate vs. time graph. The temperature was kept constant throughout.

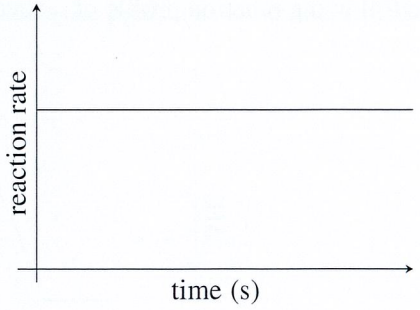
Question 6

Which of the following graphs best represents the rate of the **forward** reaction over time?

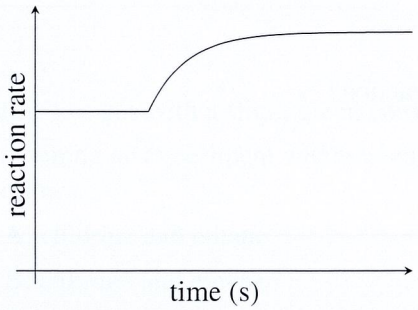
A.



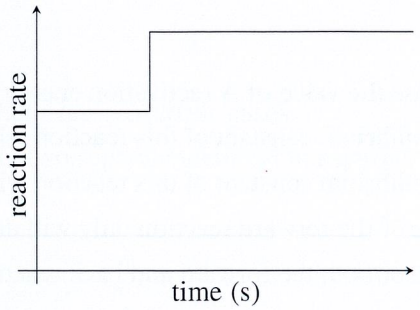
B.



C.



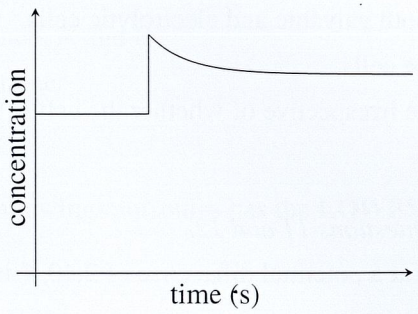
D.



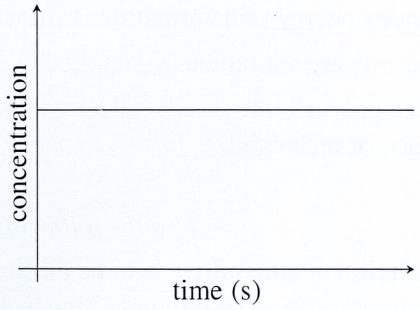
Question 7

Which of the following graphs best represents the concentration of **hydrogen** gas over time?

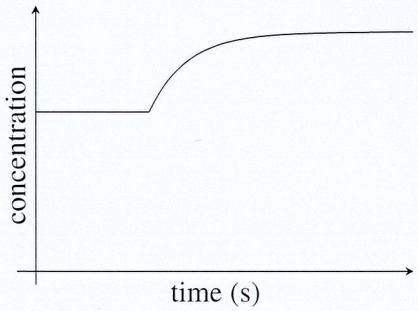
A.



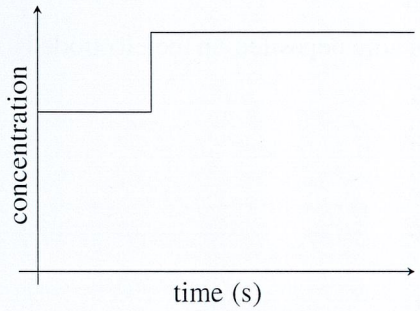
B.



C.



D.



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Question 8

Which of the following is the **best** expression for the function of a catalyst?

A. It shifts the position of equilibrium forward in a reversible reaction.

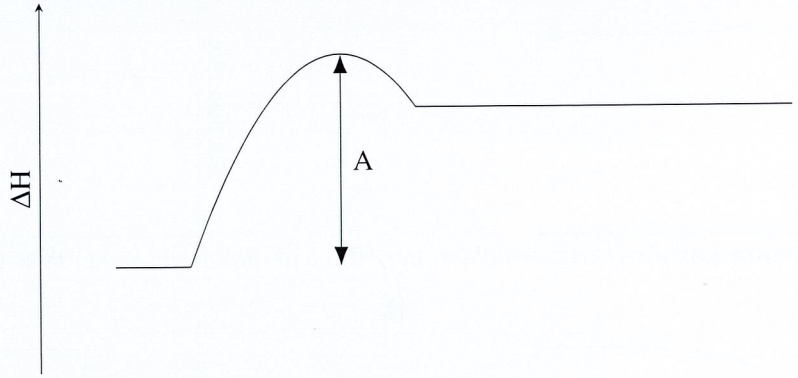
B. It increases the rate of reaction by offering an alternative reaction pathway with a lower activation energy.

C. It increases the rate of reaction by lowering the activation energy of the reaction.

D. It increases the spontaneity of a reaction.

Question 9

Consider the following reaction profile of a **reversible** reaction.



If we increase the value of A (activation energy), what will occur to the reaction?

A. The equilibrium constant of this reaction will increase.

B. The equilibrium constant of this reaction will decrease.

C. The rate of the forward reaction only will decrease.

D. The rate of both the forward and back reactions will decrease.

Question 10

Which of the following statements is **TRUE**?

A. The cations in the electrolyte or salt bridge flow towards the positive electrode in both galvanic and electrolytic cells.

B. The primary energy transformation is from electrical to chemical in both galvanic and electrolytic cells.

C. The reactants are continuously supplied in all galvanic and electrolytic cells.

D. In a secondary cell, the positive electrode is always the same electrode irrespective of whether the cell is being discharged or recharged.

Use the following information to answer Questions 11 and 12.

A 1.00 M solution of zinc nitrate is to be electrolysed. A 0.839 A current at a potential difference of 2.40 V is passed through the solution for 10.0 minutes. Zinc is deposited on one electrode.

Question 11

The mass of zinc deposited on the electrode is

A. 0.341 g.

B. 0.166 g.

C. 0.171 g.

D. 0.331 g.

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Question 12

This mass of zinc is deposited on the

A. cathode, which is the positive electrode.

B. cathode, which is the negative electrode.

C. anode, which is the positive electrode.

D. anode, which is the negative electrode.

Question 13

A 1.00 M solution of manganese (II) iodide is to be electrolysed. Which products are predicted to be released at each electrode?

	positive electrode	negative electrode
A.	iodine	hydrogen
B.	oxygen	manganese
C.	iodine	manganese
D.	oxygen	hydrogen

Question 14

Elemental chlorine is a gas with a slight green colour, whereas ethene and ethene are colourless gases.

A student is performing an experiment where he mixes some gases together in two separate identical transparent containers, as follows:

- **Container A**: chlorine and ethane
- **Container B**: chlorine and ethene

Both containers are placed into the sunlight for a few hours.

In which one or more containers do the contents decolourise?

A. container A

B. container B

C. both containers A and B

D. neither container

Question 15

Which of the following molecules has the **LOWEST** boiling point?

A. octane

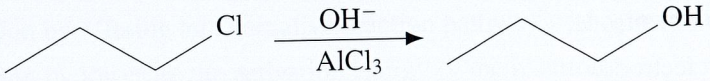
B. oct-5-ene

C. octanoic acid

D. nonane

Use the following information to answer Questions 16 and 17.

1-chloropropane can be converted to propan-1-ol via the addition of hydroxide ions, using an $AlCl_3$ catalyst. The process is depicted below.



Question 16

This reaction can be classified as

A. addition.

B. substitution and redox.

C. substitution.

D. addition and redox.

Question 17

The atom economy of this reaction is

A. 26.2%.

B. 62.8%.

C. 76.4%.

D. 100%.

Question 18

A freshly synthesised organic compound is to have its structure confirmed.

Which of the following techniques would be least appropriate in determining the structure of the compound?

A. mass spectrometry

B. UV-visible spectroscopy

C. infra-red spectroscopy

D. nuclear magnetic resonance spectroscopy

Use the following information to answer Questions 19 and 20.

The NMR spectrum of an organic compound with molecular formula $C_3H_6O_2$ has a triplet at 1.3 ppm, a quartet at 4.2 ppm and a singlet at 8 ppm.

Question 19

This organic compound is

A. acidic.

B. basic.

C. neither acidic nor basic.

D. amphiprotic.

Question 20

The IR spectrum of this peak is expected to have a strong absorption band at approximately

A. 1700 cm^{-1}.

B. 1900 cm^{-1}.

C. 2100 cm^{-1}.

D. 2300 cm^{-1}.

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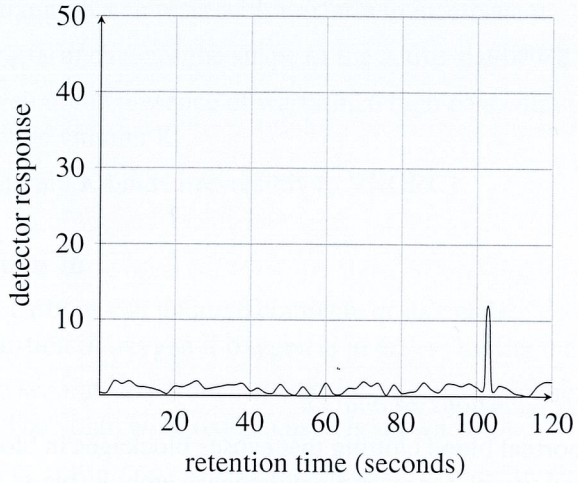
Question 21

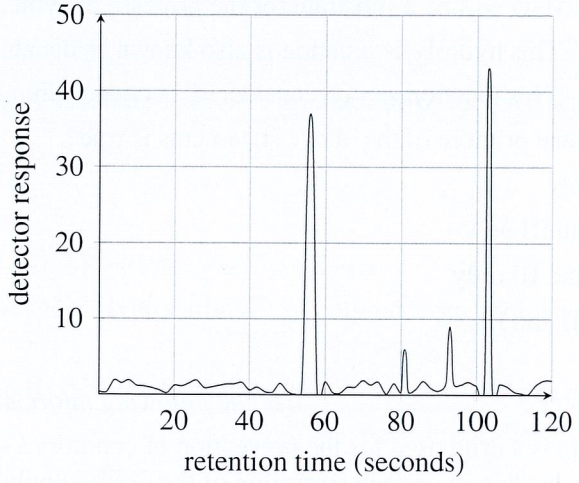
Caffeine is a compound that is found in soft drinks. The quantity of caffeine in a set of different soft drinks is to be determined via high-performance liquid chromatography.

Below are the chromatograms of a sample of pure caffeine (50 ppm) and three soft drinks.

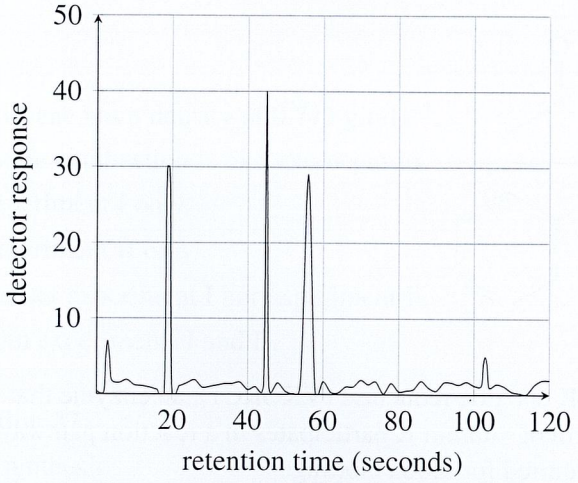
Caffeine standard (50 ppm)



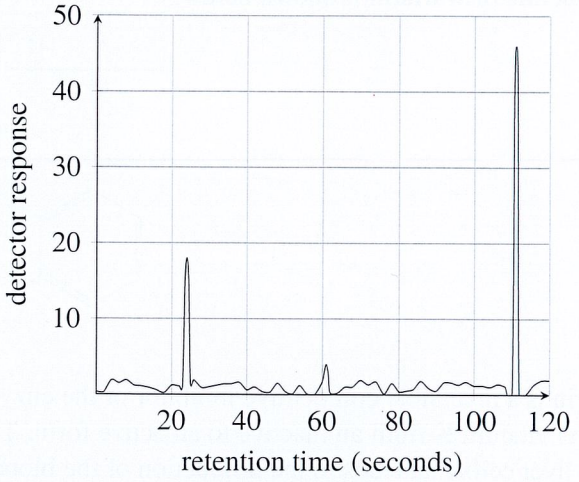
Soft drink B



Soft drink A



Soft drink C



Which of the following has the highest caffeine content?

A. soft drink A

B. soft drink B

C. soft drink C

D. a **150** ppm caffeine standard

Question 22

The interactions between deprotonated carboxy groups ($RCOO^-$) and protonated amino groups (RNH_3^+) in different R-groups of a protein are called

A. hydrogen bonding.

B. ionic interactions.

C. covalent interactions.

D. hydrophobic interactions.

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Question 23

Proteases are enzymes that hydrolyse proteins into their constituent amino acids. These are secreted by the pancreas in humans after the ingestion of food. When the food enters the small bowel, the proteases released by the pancreas hydrolyse proteins into their constituent amino acids.

Consider the following statements:

I H_2O will be a substrate for the protease enzyme.

II This hydrolysis reaction is also known as denaturation of the protein.

III This reaction can be considered a condensation reaction.

Which one or more of the above statements is true?

A. I only

B. I and III only

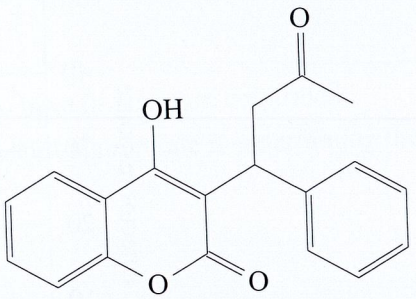
C. II and III only

D. I, III and III

Use the following information to answer Questions 24 and 25.

Warfarin is a drug used for the prevention of conditions involving abnormal blood clotting that causes blockages in blood vessels, leading to oxygen starvation of the tissue supplied by the blood vessel, for example pulmonary emboli (blood clots in the arteries supplying the lungs) and strokes (blood clots in the arteries supplying the brain).

The structure of warfarin is shown below:



warfarin

Warfarin is a reversible competitive inhibitor of the enzyme vitamin K epoxide reductase (VKORC1), an enzyme that converts vitamin K from an inactive to an active form. The active form of vitamin K participates in a reaction pathway within liver cells that leads to the production of the blood proteins required for blood clotting.

Question 24

Consider the following functional groups:

I ester

II ketone

III hydroxyl

Which of the above functional groups are found in warfarin?

A. II only

B. III only

C. II and III only

D. I, II and III

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Question 25

A warfarin overdose can cause excessive inhibition of blood clotting, which can potentially lead to life-threatening bleeding. It is known that a high dose of vitamin K given intravenously can be used as an antidote against a warfarin overdose.

The reason that high-dose vitamin K can be used as an antidote is because

A. vitamin K can react with warfarin to inactivate it.

B. warfarin changes the shape of the active site of VKORC1.

C. even in the presence of warfarin, a high concentration of vitamin K substrate leads to a greater rate of the production of active vitamin K.

D. vitamin K binds irreversibly to VKORC1.

Question 26

Hydrocarbons can undergo complete or incomplete combustion in oxygen. Hydrocarbons can only undergo complete combustion in oxygen if oxygen is in excess during the reaction.

In two separate experiments, 5.00 mL of pure liquid hexadecane ($C_{16}H_{34}$) is combusted in fixed volumes of oxygen gas at SLC. The volumes of oxygen used is shown below:

Experiment	Volume of oxygen (L)
I	12.0
II	15.0

Hexadecane has a density of 0.773 g mL^{-1}.

Complete combustion is likely to occur in

A. experiment I only.

B. experiment II only.

C. neither experiment I nor experiment II.

D. both experiments I and II.

Question 27

Photosynthesis

A. involves the production of light energy by plants.

B. utilises water as a reactant.

C. leads to the production of greenhouse gases.

D. involves the oxidation of carbon dioxide.

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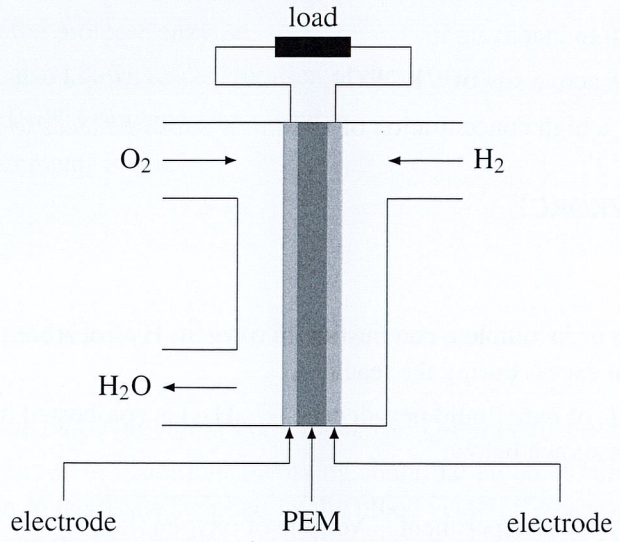
Use the following information to answer Questions 28 and 29.

The use of hydrogen as a fuel for buses has been trialled in the previous decade.

The hydrogen in these buses is utilised in a stack of fuel cells with a polymer electrolyte membrane used as an electrolyte. A simplified diagram of one of these fuel cells is shown below:



These buses are fitted with hydrogen tanks that could be refilled regularly at filling stations. The hydrogen gas could be sourced industrially from the steam reformation of methane derived from fossil fuel sources via the following reaction:

$$CH_4\ (g) + 2\ H_2O\ (g) \rightleftharpoons CO_2\ (g) + 4\ H_2\ (g)$$

The fuel cell in a particular hydrogen-powered bus has a maximum power of 100 kW, in which the current generated by the fuel cell is 350 amperes.

Question 28

What is the maximum hydrogen fuel consumption by this bus, in litres (at SLC) per hour?

A. 162

B. 648

C. 324

D. 6.53

Question 29

It is claimed that hydrogen-powered buses are more environmentally friendly than the use of conventional diesel-powered buses that use an internal combustion engine to combust diesel in oxygen gas.

A suitable argument that would support this claim would be that

A. there would be no harmful greenhouse gas emissions caused by the widescale utilisation of hydrogen-powered buses.

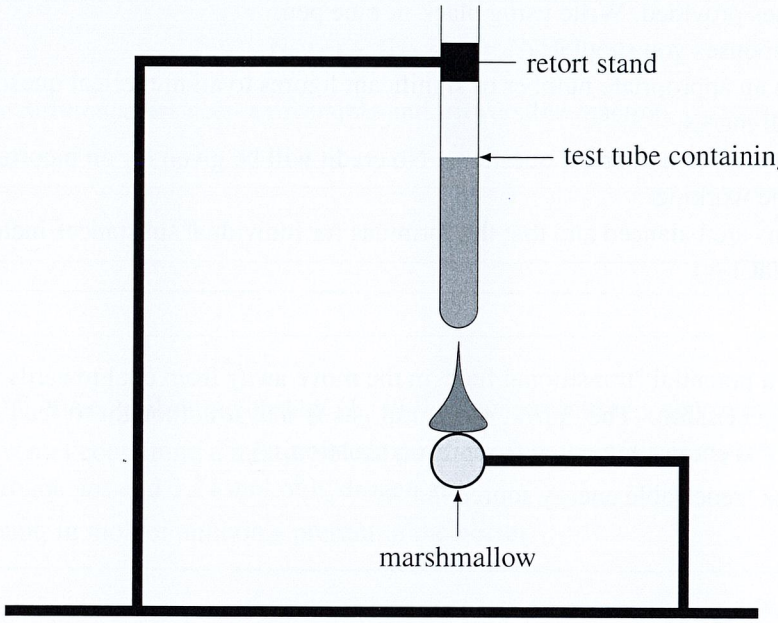
B. water vapour is the only exhaust gas.

C. there would be fewer harmful greenhouse gas emissions caused by the widescale utilisation of these buses due to the greater energy efficiency of the hydrogen fuel cell.

D. hydrogen is a renewable fuel.

Question 30

A 0.0329 g piece of marshmallow was ignited and held against a test tube containing 10.0 mL of water. A diagram of the experimental setup is shown below:



Marshmallows are known to be composed almost entirely of carbohydrate. The energy content of carbohydrates can be estimated to be 16 kJ g^{-1}.

In this experiment, the temperature of the water increased from 19.5°C to 27.6°C.

The percentage heat loss to the surroundings is closest to

A. 35%.

B. 45%.

C. 55%.

D. 65%.

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SECTION B - Short answer questions

Instructions for Section B

Answer **all** questions in the spaces provided. Write using black or blue pen.

To obtain full marks for your responses you should

- give simplified answers with an appropriate number of significant figures to all numerical questions; unsimplified answers will not be given full marks.
- show all working in your answers to numerical questions. No credit will be given for an incorrect answer unless it is accompanied by details of the working.
- make sure chemical equations are balanced and that the formulas for individual substances include an indication of state; for example, H_2 (g); NaCl (s)

Question 1 (7 marks)

Natural gas is being discussed as a potential "transitional fuel" in the move away from coal towards renewable sources of energy for electricity generation. The appeal of natural gas is that it is thought to lead to less greenhouse gas emissions per unit of energy generated from its combustion.

a. What is meant by the phrase "renewable energy source"? 1 mark

The energy content of pure methane and carbon are 55.0 kJ g^{-1} and 33.0 kJ g^{-1} respectively.

b. Write down balanced combustion equations for the **complete** combustion of: 2 marks

methane

carbon

c. Calculate the volume of carbon dioxide produced at SLC per MJ of energy released when each of **natural gas** and **coal** is combusted, and explain whether your answer supports the premise for using natural gas as a transitional fuel. 4 marks

You may assume natural gas is composed of pure methane and that coal is composed of pure carbon.

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Question 2 (13 marks)

Ammonia (NH_3) is a highly pungent gas at room temperature with a wide variety of uses including in fertilisers.

Ammonia is produced via the Haber process, which consists of a single reversible reaction as written below:

$$N_2\,(g) + 3H_2\,(g) \rightleftharpoons 2NH_3\,(g)$$

a. Briefly explain the difference between a reversible and irreversible reaction. 1 mark

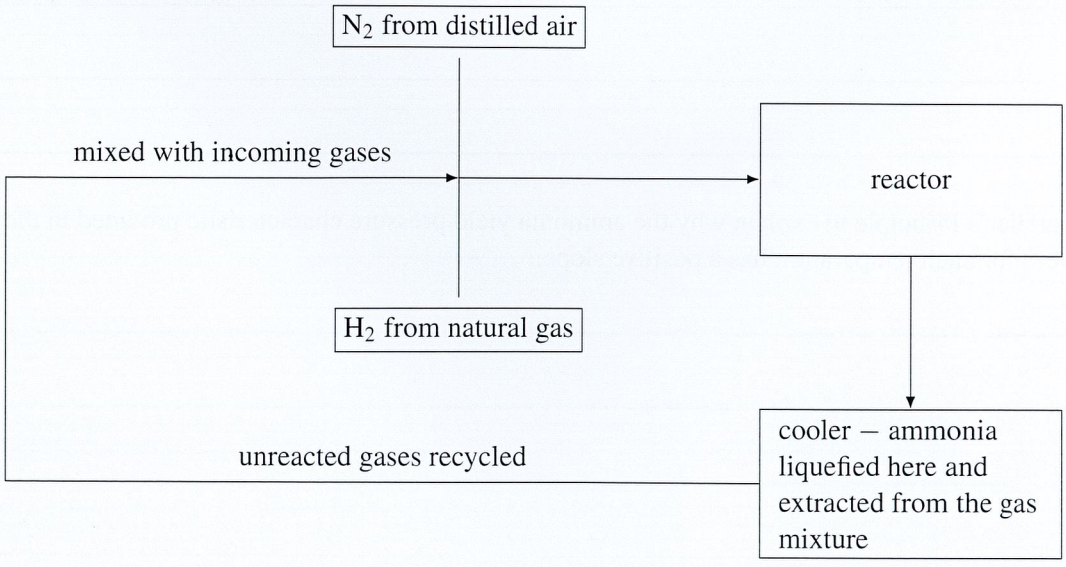
At a temperature of 470°C, the equilibrium constant (K) of the above reaction is approximately 1.18×10^{-1}.

Consider a 20.0 L rigid vessel containing a mixture of all three gases at equilibrium. It is known that the vessel contains 1.05 mol of nitrogen gas and 3.24 mol of hydrogen gas.

b. Calculate the amount, in mol, of ammonia present in the vessel. 2 marks

In the Haber process, the nitrogen gas is sourced from the fractional distillation of liquefied air, and the hydrogen gas is sourced from the steam reformation of natural gas. The two gases are added in their stoichiometric ratios.

A schematic for the Haber process is shown below:



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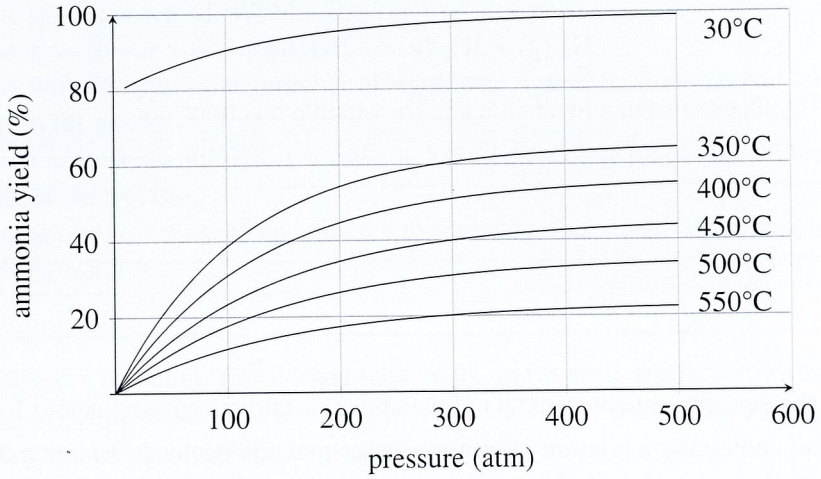
The ammonia yield-pressure characteristic at various temperatures is shown below (assuming a 1:3 mole ratio of added nitrogen and hydrogen gases and no reactant recycling).



c. Use equilibrium principles to explain the impact of the following on the yield of ammonia produced.

i. the actions of the cooler 1 mark

ii. mixture of recycled gases with incoming gases 1 mark

iii. the presence of a metal catalyst in the reactor 1 mark

d. Use Le Chatelier's Principle to explain why the ammonia yield pressure characteristic provided in the question stem for each temperature has a positive slope. 2 marks

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e. What information about the enthalpy of reaction can be determined by the information provided? Explain your answer. 2 marks

In industry, a temperature of approximately 400 °C and pressure of 200 atm are used.

f. Explain why higher temperatures and pressures are not used in the Haber process. 2 marks

Scientists are researching and trialling other catalysts, including enzymes, that could be used in the Haber process. It is thought that an enzymatic catalyst, at room temperature, would dramatically increase the reaction rate to an extent that higher pressures and temperatures cannot.

g. Use the ammonia yield pressure characteristic on the previous page to explain how an enzymatic catalyst could, directly or indirectly, facilitate a greater **yield** of ammonia. 1 mark

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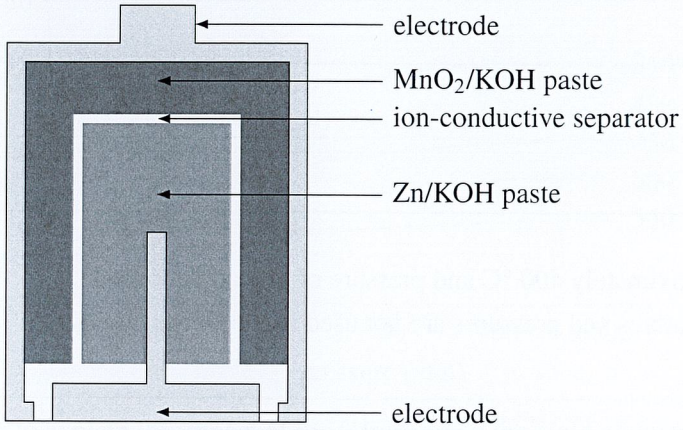
Question 3 (3 marks)

The alkaline cell is a commercial primary cell in which elemental zinc is converted to zinc(II) oxide and manganese dioxide (MnO_2) is converted to manganese oxyhydroxide (MnO(OH)), with the electrolyte consisting of a concentrated paste of 7 M KOH, which is mixed with manganese dioxide.

The structure of this cell is as shown below:



a. Write the half-equation of the reaction that occurs at the **positive electrode**. 1 mark

b. State the major energy conversion that occurs in this cell as it is being discharged. 1 mark

c. Draw an arrow to depict the direction of flow of potassium ions in this cell. 1 mark

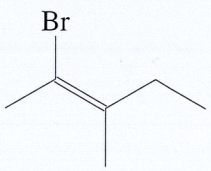
Question 4 (4 marks)

a. Draw the structure of the following compounds in the boxes provided.

i. 3-chloro-butan-1-amine 1 mark

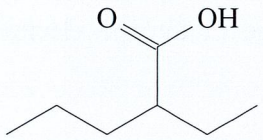
ii. 3-methylbutan-2-ol 1 mark

b. Write down the IUPAC systematic names for the following compounds:

i. 

1 mark

ii.

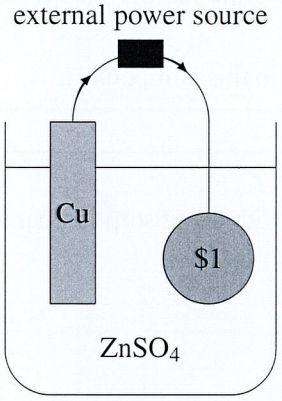


1 mark

Question 5 (8 marks)

Jin found an old $1 coin while cleaning her desk. She decided to use this $1 coin to investigate the possibility of restoring the coin by electroplating a new layer of metal on it.

She does an online literature search and learns that coins are mostly made out of zinc and copper, so she decided to use a copper electrode and a solution of zinc sulfate, as per the following setup:



The current is 1.00 amperes and the potential difference is 5 volts.

a. Write the half-equation of the reaction that initially occurs at the **negative** electrode. 1 mark

b. Determine the change in mass of the anode per second. 3 marks

In this cell, if the potential difference applied and therefore the current that passes through the cell were too high, side reactions can occur at each electrode.

c. Write down the half-equations of the **TWO** reactions that would occur at the **anode** if the potential difference applied across the cell were too high. 2 marks

d. How would the composition of the coin change if the copper electrode were replaced with a platinum electrode? Explain your answer. 2 marks

Question 6 (9 marks)

The IR, ^{1}H NMR and mass spectra of a particular compound is shown on the next page. This compound consists of 54.5% C, 9.1% H and 36.4% O by mass.

a. How many hydrogen environments are there in this compound? 1 mark

b. Identify the **TWO** bonds that are associated with the absorption band labelled **X** on the infrared spectrum. 2 marks

c. Write down the m/z value of the **base peak** on the mass spectrum. 1 mark

d. Use the data provided to determine the **empirical** formula of this compound. 1 mark

e. Determine the **molecular** formula of compound **A**. 1 mark

The chemical environment corresponding to the peak at 11.9 ppm in the ^{1}H NMR spectrum for this compound contains 1 hydrogen atom.

f. How many H atoms are there in the chemical environment corresponding to the doublet on the ^{1}H NMR spectrum? 1 mark

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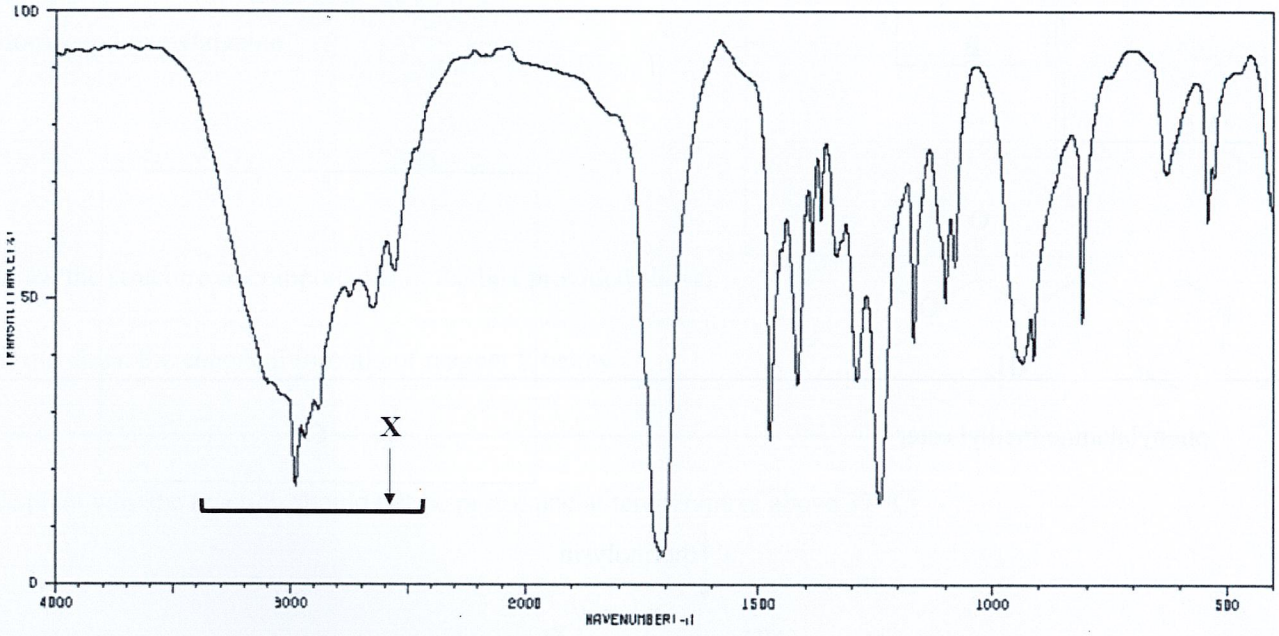
g. Draw the structure of the organic compound based on the data provided. 1 mark

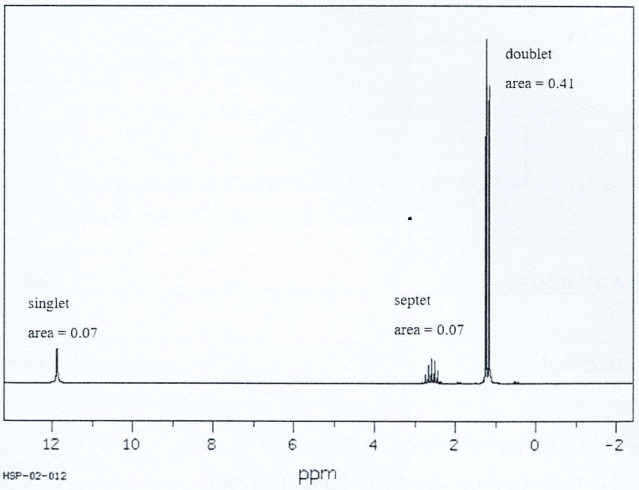
h. Write down the IUPAC systematic name for the compound. 1 mark

_____

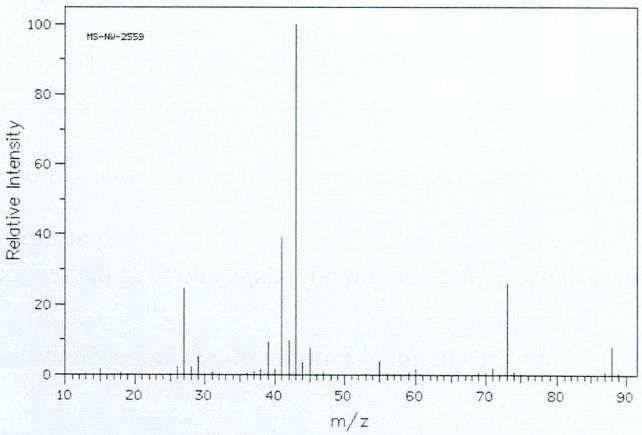
IR spectrum



^{1}H NMR spectrum



Mass spectrum



(Source: Spectral Database for Organic Compounds, National Institute of Advanced Industrial Science and Technology, Japan)

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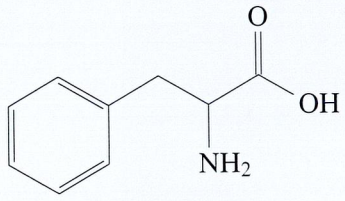
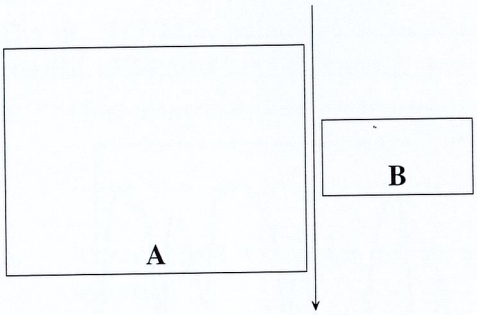
Question 7 (9 marks)

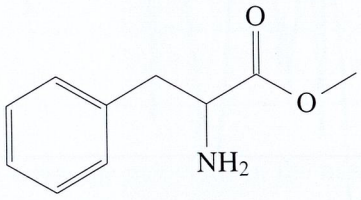
Aspartame is an artificial peptide-based sweetener that is sold commercially as Equal ™. It is a modified dipeptide of phenylalanine and a second amino acid.

A simplified reaction mechanism for the synthesis of aspartame is shown below.



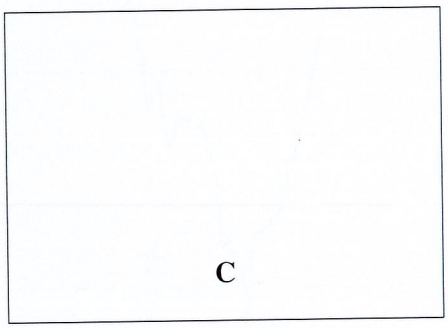
phenylalanine



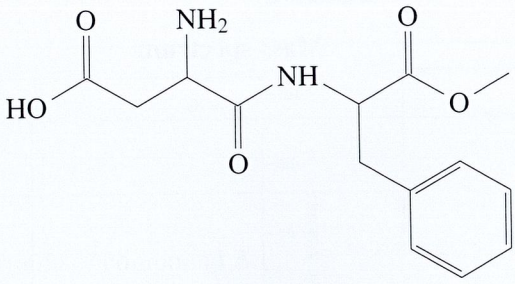


phenylalanine methyl ester

+



thermolysin



aspartame

a. Draw the structures of compounds **A** and **C** in the reaction pathway above. 1 mark

b. Write down the chemical formula of a suitable catalyst in the box labelled **B**. 1 mark

The synthesis of aspartame requires the use of a specific enantiomer of phenylalanine, referred to as L-phenylalanine. Often, however, phenylalanine is supplied as a 50:50 mixture of the two enantiomers, L-phenylalanine and D-phenylalanine. This is known as a racemic mixture.

c. Circle the chiral centre on the phenylalanine molecule on the above diagram. 1 mark

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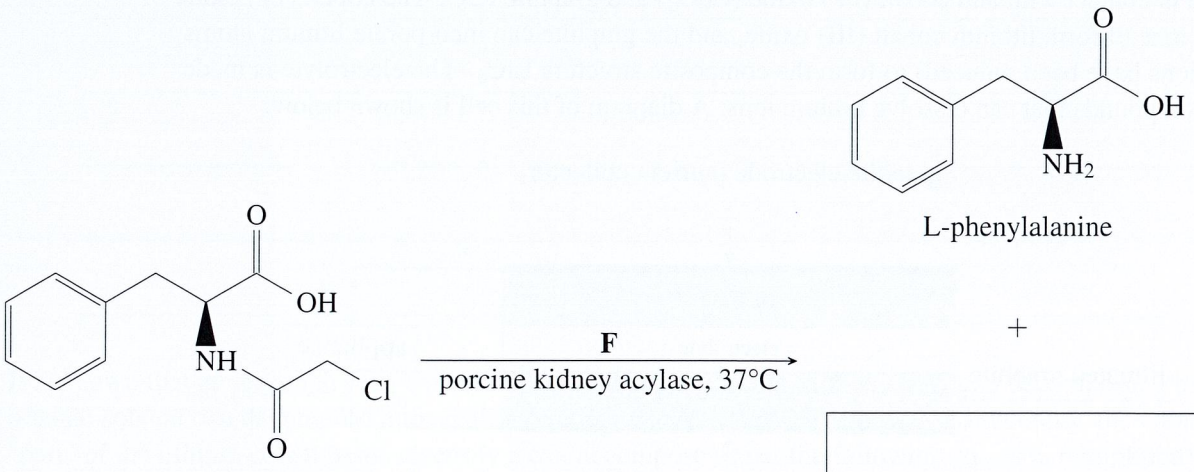
The mixture of D- and L-phenylalanine is treated to form a mixture of D- and L-chloroacetylphenylalanine (structure shown below). The product is subsequently mixed with a solution of porcine kidney-derived acylase to form L-phenylalanine in the hydrolysis reaction shown below:



d. Draw the structure of compound **E** in the box provided above. 1 mark

e. Write down the chemical formula of reagent **F** below. 1 mark

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f. Explain why the reaction should not be performed at temperatures above 37 °C. 2 marks

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g. Explain, at a molecular level, why after mixing with the solution of porcine kidney acylase, only the L-enantiomer was produced. 2 marks

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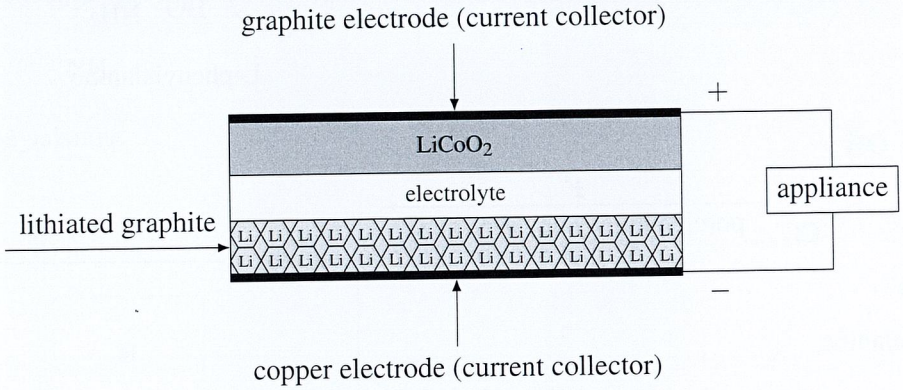
Question 8 (10 marks)

Lithium ion batteries are secondary cells that are used in high-powered devices such as mobile phones.

The two electrodes are in contact with and cobalt (IV) oxide (CoO_2) and graphite (C_6). The cobalt (IV) oxide can react with lithium ions to form lithium cobalt (III) oxide, and the graphite can incorporate lithium atoms (i.e. once the lithium ions have been reduced) to form the composite structure LiC_6. The electrolyte is made out of liquid organic compounds that can dissolve lithium ions. A diagram of this cell is shown below:



During **discharge**, the set of reactions that occur at each electrode is shown below (simplified):

$$LiC_6 \rightarrow Li^+ + e^- + C_6$$
$$CoO_2 + Li^+ + e^- \rightarrow LiCoO_2$$

a. Write down the reaction equation that occurs at the **positive** electrode during **recharge**. 1 mark

_____

b. Explain why an aqueous electrolyte is not used. Include a reaction equation in your answer. 2 marks

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If too high a current is applied to recharge the cell, metallic lithium can be plated on the surface of graphite electrode, instead of being intercalated within the graphite as LiC_6. The plating can be **dendritic** in nature, with finger-like projections emanating from the graphite electrode. The danger is that one of these finger-like projections can come into contact with the $LiCoO_2$ electrode.

This can lead to a condition called **thermal runaway**, where the temperature inside the battery increases at an ever-increasing rate.

c. Explain how an increase in temperature increases reaction rate, using collision theory. 2 marks

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d. State the energy transformation that occurs when the dendritic Li plate contacts the other electrode, and therefore explain how this can lead to thermal runaway. 2 marks

It is known that at very high temperatures some of the components of the battery can decompose. Firstly, the organic solvent can decompose into small hydrocarbon molecules (eg. ethane) and secondly, the cobalt oxide "part" of the lithium cobalt oxide electrolyte can decompose as per the following equation (simplified):

$$3\ CoO_2 \rightarrow Co_3O_4 + O_2$$

There have been instances where lithium ion batteries have exploded in devices such as mobile phones, aeroplanes and electric cars, which can cause serious injury to the owners or occupants of such devices.

e. Explain why using too high a current to recharge such a phone can lead to an explosion. Use an equation in your answer. 3 marks

Question 9 (12 marks)

Methanol is a liquid with a wide variety of uses in industry. It can be synthesised industrially from fossil fuel and biofuel sources.

Methanol is known to be less viscous than butan-1-ol.

a. Explain what is meant by the term "viscosity" and explain the difference in viscosities between methanol and butan-1-ol. 2 marks

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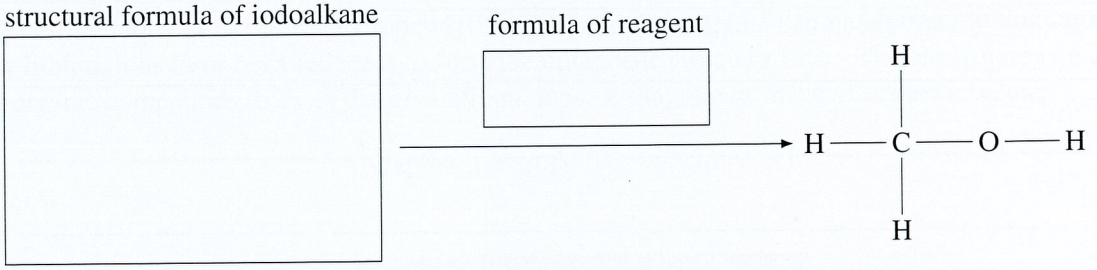
b. Methanol could theoretically be produced from a liquid iodoalkane in a single-step reaction. 1 mark
Fill in the blanks below to complete the equation for this reaction.



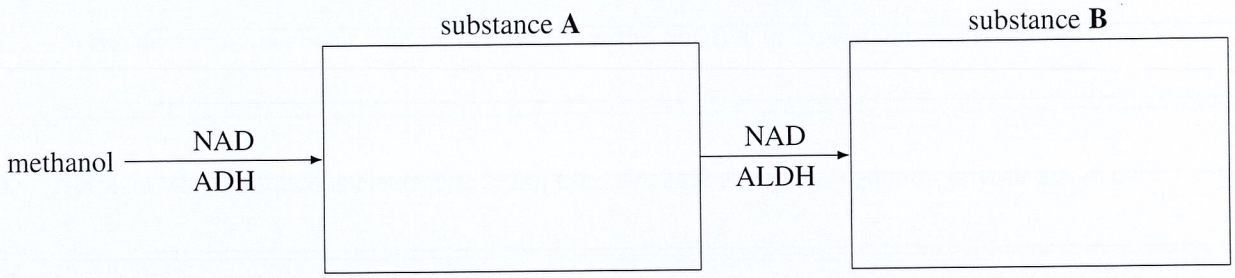
Methanol poisoning is a clinical problem that, not uncommonly, presents to Emergency departments. Ultimately, the primary causes of methanol poisoning are the oxidation products of methanol.

In humans, there are two enzymes, namely ADH and ALDH, that catalyse the oxidation reaction of methanol and its intermediate products. Also, human cells contain significant quantities of NAD (nicotinamide adenine dinucleotide) which is an oxidising agent that acts as an additional substrate for ADH and ALDH.

Methanol is initially oxidised in the presence of ADH to produce substance **A**, which itself is subsequently oxidised in the presence of ALDH to produce substance **B**. It is known that substances **A** and **B** are both liquids at room temperature and if substance **B** were dissolved in a solution of sodium bicarbonate ($NaHCO_3$), bubbling would be observable.

It is substance **B** which is most toxic to humans, causing blindness among other clinical problems.

A partially filled diagram depicting the above information is shown below:



c. Deduce the structures of **A** and **B** and draw them in the boxes provided above, **showing all bonds**. 2 marks

It is known that ADH and ALDH have an optimum temperature in which they function.

d. Explain why the activities of ADH and ALDH deteriorate when the temperature increases or decreases from the optimum temperature. 2 marks

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Methanol is also a reagent in the production of biodiesel, which is classed as a biofuel.

e. Explain why biodiesel can be classed as a biofuel. 1 mark

f. Linolenic acid is a fatty acid with chemical formula $C_{17}H_{29}COOH$. 1 mark
Write a balanced equation for the production of a biodiesel molecule using methanol and linolenic acid as the starting reagents.

Siobhan, a VCE Chemistry student, is given an unlabelled bottle containing liquid methanol. She wishes to perform an experiment whose aim is to determine whether the methanol is contaminated with small quantities of ethanol and to quantify the degree of contamination.

She proposes to use high performance liquid chromatography for this experiment. Her classmate, Claire, suggests using mass spectrometry instead.

g. Compare the analytical techniques suggested by Claire and Siobhan with respect to their suitability in achieving Siobhan's aim in her experiment. 3 marks

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Question 10 (15 marks)

A VCE Chemistry 3/4 student has decided to perform an investigation entitled *Volumetric determination of the ethanol content in Grey Goose vodka*. Below is an excerpt of his scientific poster.

Volumetric determination of the ethanol content in Grey Goose vodka

Introduction

Vodka is an alcoholic beverage that is derived from the fermentation of starch from potatoes and cereal grains. It is highly popular particularly amongst young adolescents, to the point that vodka-containing alcoholic mixed drinks, colloquially known as "alcopops", are in high demand in the market. Grey Goose is a particular brand of vodka from France whose alcohol content is advertised as 40.0% (v/v). This experiment aims to determine the alcohol content (present as ethanol) in a sample of Grey Goose vodka via volumetric titration against a standard solution of potassium permanganate.

Methodology

A 50.00 mL sample of vodka is transferred into a 500.0 mL volumetric flask and made up to the mark with water. From this volumetric flask, 20.00 mL aliquots were transferred into a conical flask and titrated against a 0.5022 M solution of potassium permanganate ($KMnO_4$).

The reaction equation is given below:

$$4\ MnO_4^-\ (aq) + 5\ C_2H_5OH\ (aq) + 12\ H^+\ (aq) \rightarrow 4\ Mn^{2+}\ (aq) + 5\ CH_3COOH\ (aq) + 11\ H_2O\ (l)$$

The permanganate ion (MnO_4^-) ion has an intensely purple colour while the Mn^{2+} ion is colourless.

Results

The titre volumes obtained are displayed in the following table.

Run	1	2	3	4	5
Titre (mL)	23.22	23.20	24.01	23.21	23.22

The average of the concordant titres is 23.22 mL.

a. Write down the half-equation of the **reduction** reaction that occured in this titration. 1 mark

b. Based on the average concordant titre, calculate the concentration, in mol L^{-1}, of ethanol in the vodka. 2 marks

The density of ethanol is 0.789 g mL^{-1}.

c. Use your answer to part **b** to calculate the ethanol content in the vodka, in % (v/v). 2 marks

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An excerpt from the student's logbook is shown below.

05.08.2024

Volumetric determination of ethanol in vodka

Procedure:

- volume of vodka extracted - 50.00 mL
- vodka diluted to 500.0 mL
- 20.00 mL aliquot pipetted to conical flask, titrated against permanganate solution

Concentration of permanganate - 0.5022 M

Observations (chemicals): vodka clear fluid, permanganate intensely purple

Observations (titrations):

- run 1 - add first drop of titrant, contents of reaction flask turn pink immediately, then after few seconds contents turn clear. add second drop of titrant, contents turn pink and immediately turn clear. add titrant in a slow stream, contents remain a very slight pink colour, stop stream - contents turn clear immediately. bubble formation observed on surface of solution. titrant then added dropwise until colour change persistent for 10 seconds.
- run 2 - as for titration #1
- run 3 - titrant added as a stream initially. contents of reaction flask turn pink immediately, after a few seconds contents turn clear. brown solid observed on bottom of the flask. otherwise, observations as per run 1.
- run 4 and 5 - as for run 1

The student conducted some research after the experiment and has found the following useful pieces of information:

- In the presence of a powerful oxidising agent, ethanoic acid can be oxidised to carbon dioxide.
- Mn^{2+} can act as a catalyst for the reduction of MnO_4^- ions.
- MnO_2 is a brown solid.

d. Write a **discussion** and **conclusion** section for this scientific poster. In your discussion, ensure that you 10 marks

- summarise the results that were obtained
- assess the reliability and validity of the results obtained
- suggest reasons as to why the data may or may not be accurate, and explanations for interesting observations
- suggest improvements to the experimental design.

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